

Unit 2

DATA LINK LAYER AND MEDIA ACCESS

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DLC SERVICES

- The data link control (DLC) deals with procedures for communication between two adjacent nodes.
- node-to-node communication—no matter whether the link is dedicated or broadcast.
- Data link control service include

(1) Framing

(2) Flow Control

(3) Error Control

FRAMING

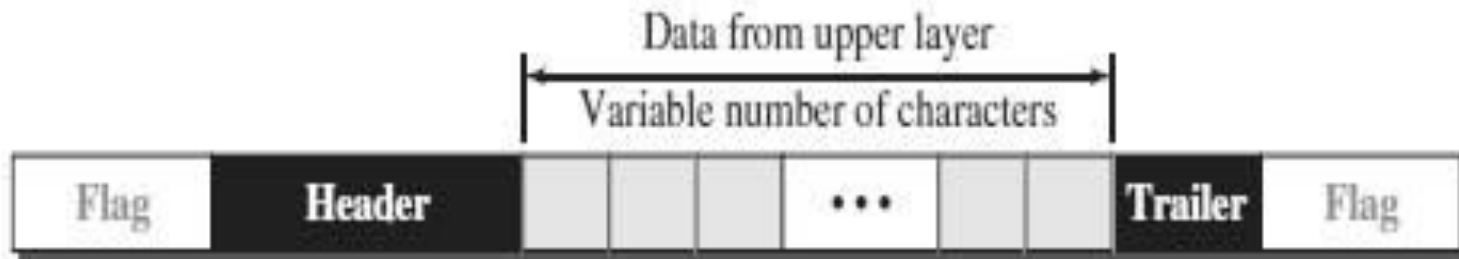
- The data-link layer packs the bits of a message into frames, so that each frame is distinguishable from another.
- When a message is carried in one very large frame, even a single-bit error would require the retransmission of the whole frame.
- When a message is divided into smaller frames, a single-bit error affects only that small frame.
- Framing in the data-link layer separates a message from one source to a destination by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

Frame Size

- Frames can be of fixed or variable size.
- Frames of fixed size are called cells. In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.
- In variable-size framing, we need a way to define the end of one frame and the beginning of the next. Two approaches were used for this purpose:
- A character-oriented approach and a bit-oriented approach

Character-Oriented Framing

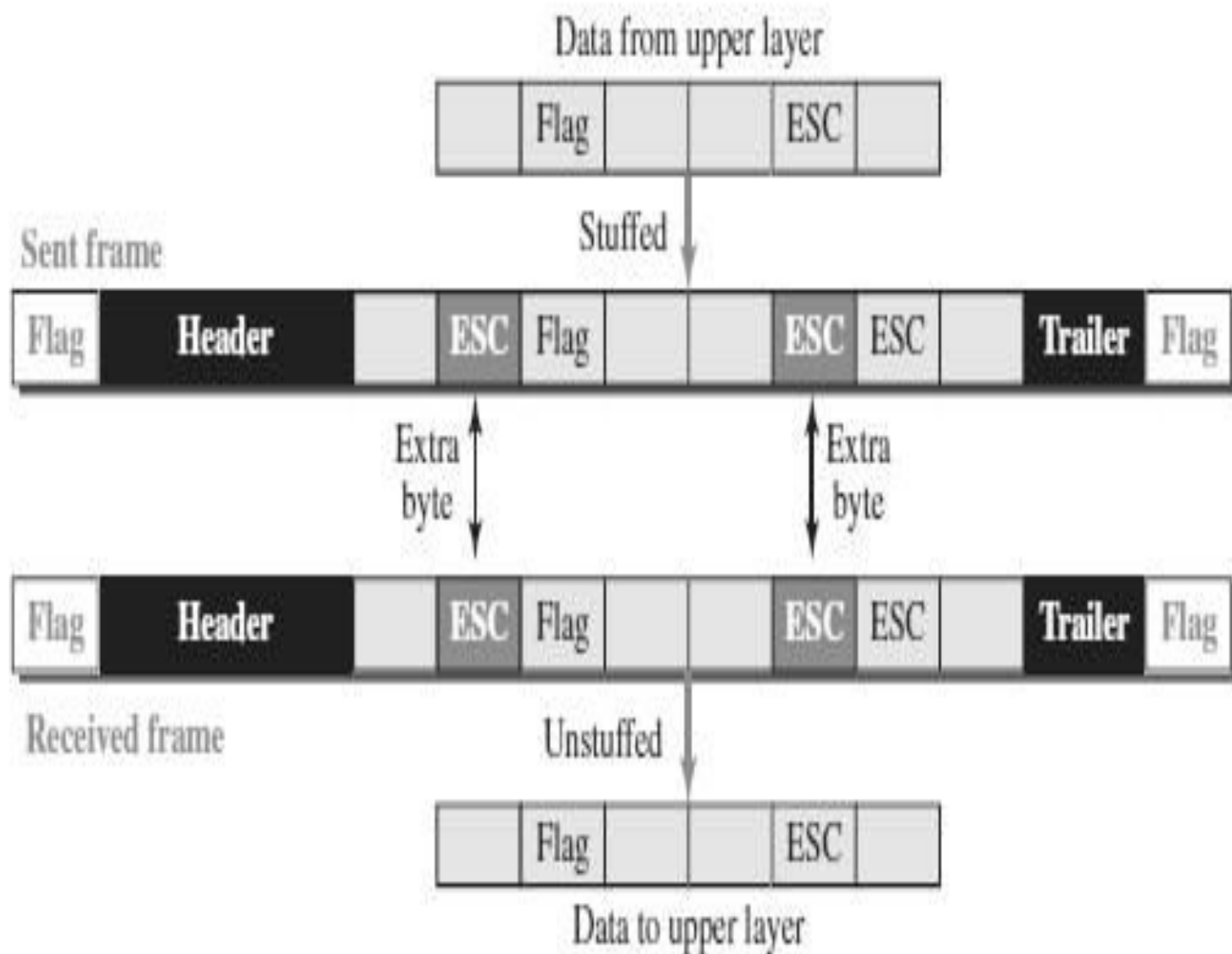
- In character-oriented (or byte-oriented) framing, data to be carried are 8-bit characters.
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame.
- The flag, composed of protocol-dependent special characters, signals the start or end of a frame.



- Any character used for the flag could also be part of the information.
- If this happens, when it encounters this pattern in the middle of the data, the receiver thinks it has reached the end of the frame.
- To fix this problem, a **byte-stuffing** strategy was added to character oriented framing.

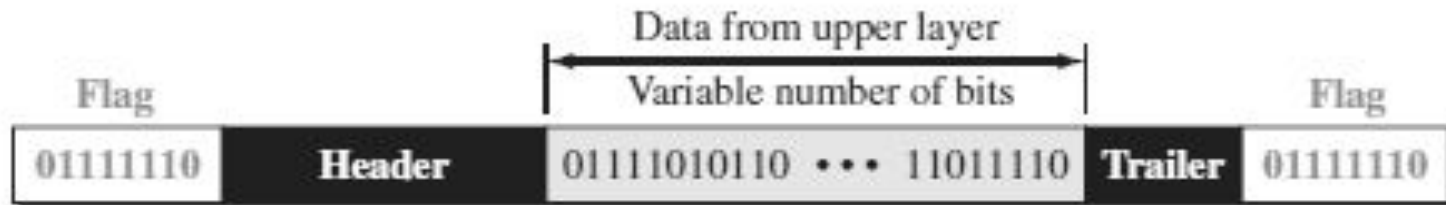
Byte Stuffing (or) Character Stuffing

- **Byte stuffing is the process of adding one extra byte whenever there is a flag or escape character in the text.**
- In byte stuffing, a special byte is added to the data section of the frame when there is a character with the same pattern as the flag.
- The data section is stuffed with an extra byte. This byte is usually called the escape character (ESC) and has a predefined bit pattern.
- Whenever the receiver encounters the ESC character, it removes it from the data section and treats the next character as data, not as a delimiting flag



Bit-Oriented Framing

- In bit-oriented framing, the data section of a frame is a sequence of bits to be interpreted by the upper layer as text, graphic, audio, video, and so on.
- In addition to headers and trailers, we still need a delimiter to separate one frame from the other.
- Most protocols use a special 8-bit pattern flag, 01111110, as the delimiter to define the beginning and the end of the frame.



- If the flag pattern appears in the data, the receiver must be informed that this is not the end of the frame.
- This is done by stuffing 1 single bit (instead of 1 byte) to prevent the pattern from looking like a flag. The strategy is called **bit stuffing**

Bit Stuffing

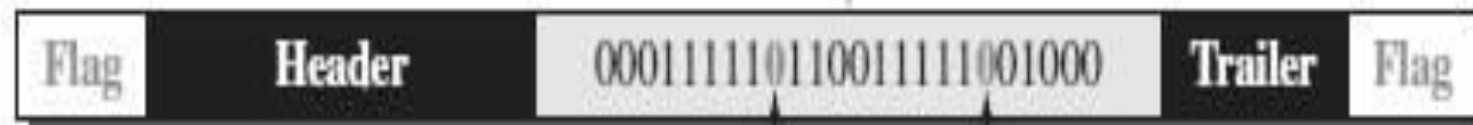
- **Bit stuffing is the process of adding one extra 0 whenever five consecutive 1s follow a 0 in the data, so that the receiver does not mistake the pattern 0111110 for a flag.**
- In bit stuffing, if a 0 and five consecutive 1 bits are encountered, an extra 0 is added.
- This extra stuffed bit is eventually removed from the data by the receiver.
- The extra bit is added after one 0 followed by five 1's regardless of the value of the next bit.
- This guarantees that the flag field sequence does not inadvertently appear in the frame.

Data from upper layer

00011111110011111101000

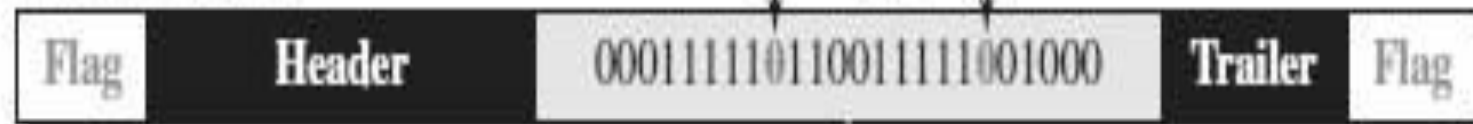
↓
Stuffed

Frame sent



↑
Two extra
bits
↑

Frame received



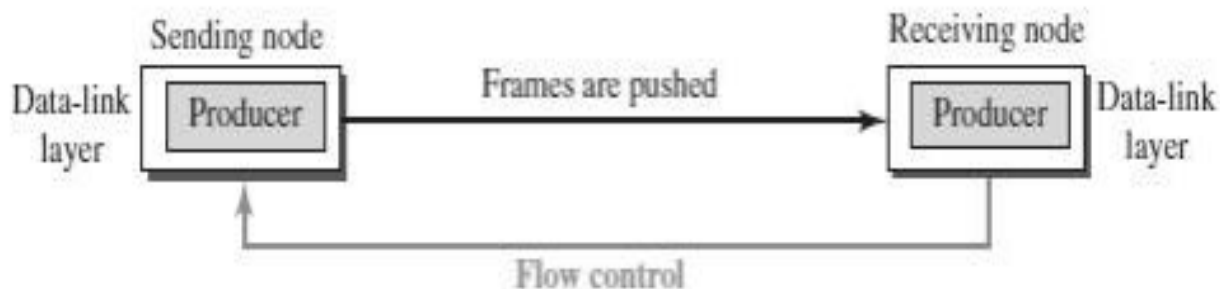
↓
Unstuffed

00011111110011111101000

Data to upper layer

FLOW CONTROL

- **Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.**
- The receiving device has limited speed and limited memory to store the data.
- Therefore, the receiving device must be able to inform the sending device to stop the transmission temporarily before the limits are reached.
- It requires a buffer, a block of memory for storing the information until they are processed.

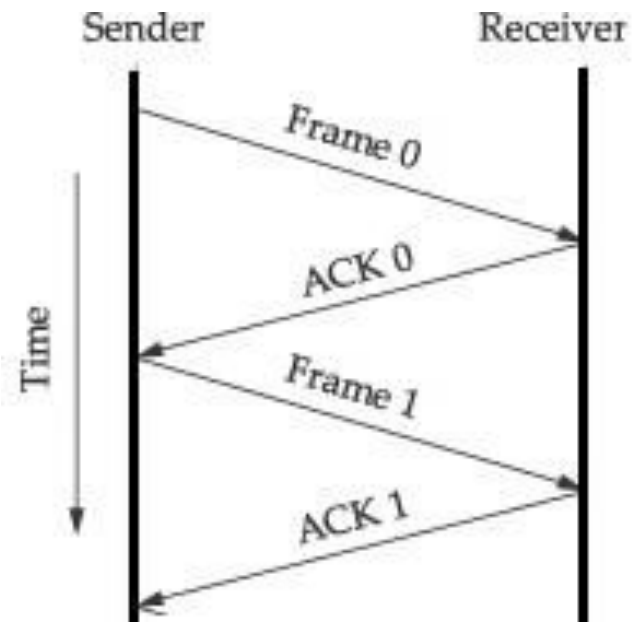
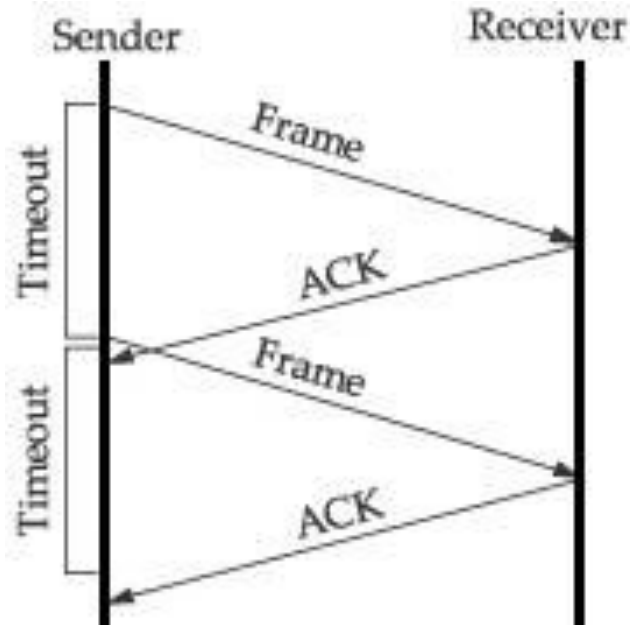
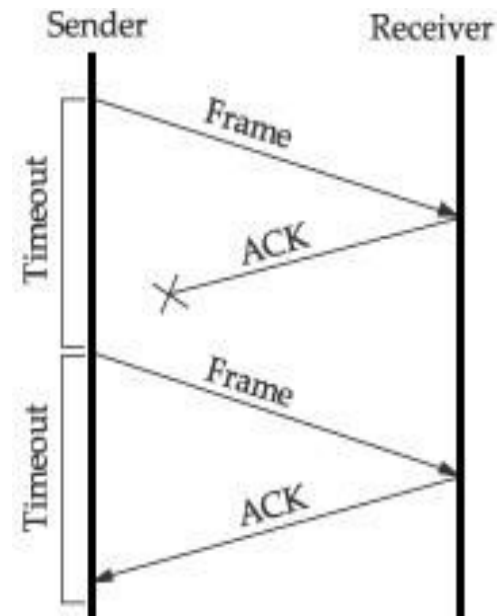
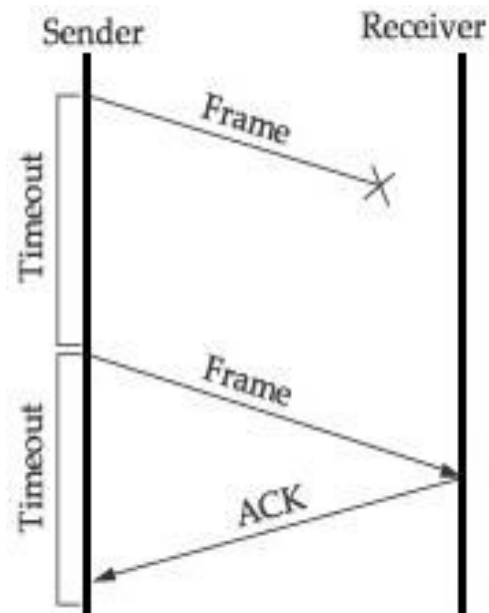
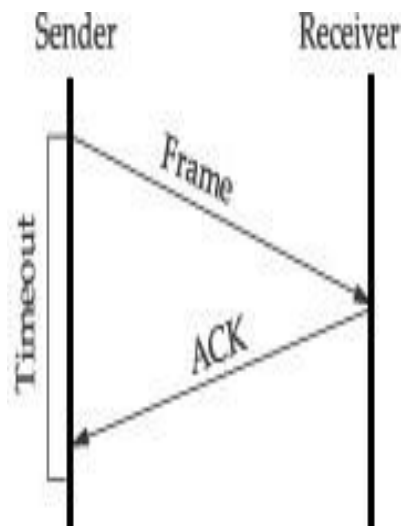


Two methods have been developed to control the flow of data:

- Stop-and-Wait
- Sliding Window

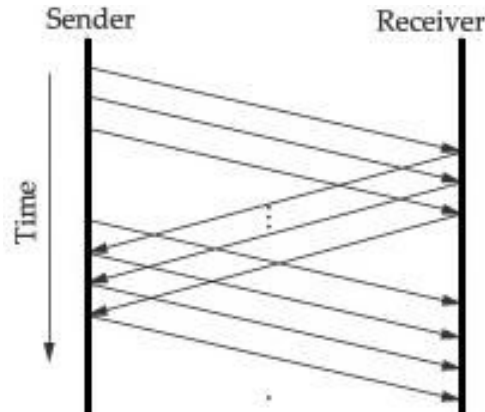
STOP-AND-WAIT

- The simplest scheme is the stop-and-wait algorithm.
- sender waits for an acknowledgement after every frame it sends.
- When acknowledgement is received, then only next frame is sent.
- The process of alternately sending and waiting of a frame continues until the sender transmits the EOT (End of transmission) frame.



2. Sliding Window

- Sliding window improves efficiency by allowing multiple frames to be in transit.



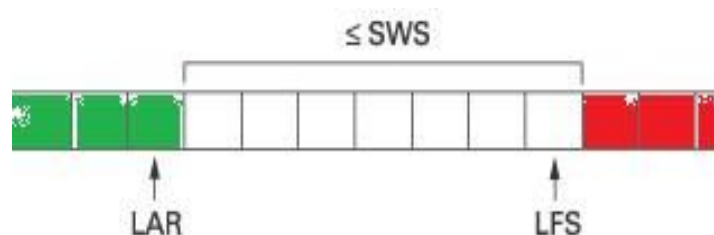
- window* defines range of sequence numbers for both sender and receiver to deal with.
- Window position changes (*slides*) due to transmission of frames and ACKs.

Sender side

- Sender assigns a sequence number SeqNum to each frame.
- A *timer* is associated with each frame it transmits, and retransmits the frame on timeout.
- It maintains three state variables:
 - (i) Sender window size SWS gives the upper bound on number of outstanding frames.

SWS is based on delay $\times$ bandwidth product.

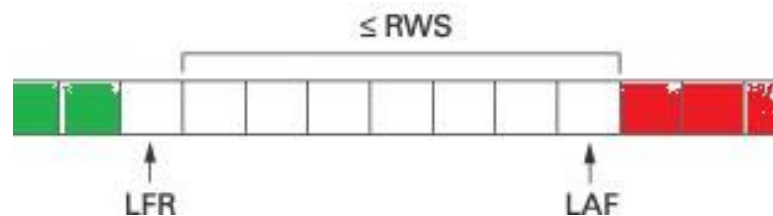
- (ii) LAR denotes the sequence number of the last acknowledgment received.
 - (iii) LFS denotes the sequence number of the last frame sent.
- Invariant $LFS - LAR \leq SWS$ is always maintained



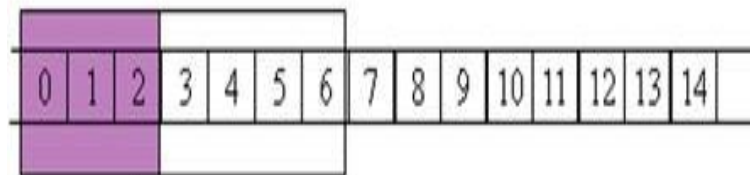
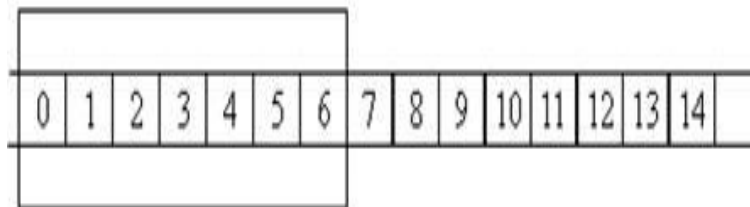
- When an acknowledgment arrives, LAR moves to the right, allowing the sender to transmit the subsequent frames.
- Sender buffers up to SWS frames (for retransmission), until they are acknowledged.

Receiver Side

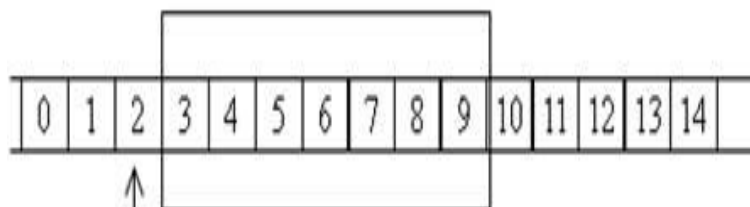
- Similarly the receiver maintains three state variables:
- Receiver window size RWS gives the upper bound on number of out-of-order frames that the receiver is willing to accept.
- LAF denotes acceptable frame with the largest sequence number
- LFR denotes sequence number of the last frame received
- Invariant $LAF - LFR \leq RWS$ is always maintained.



Sender

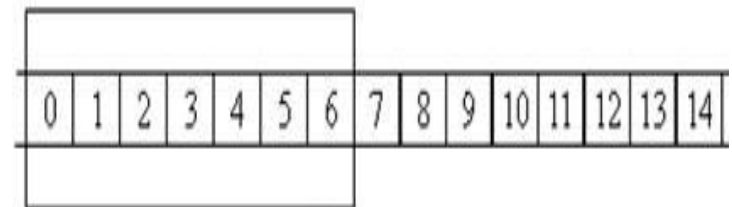


LFS

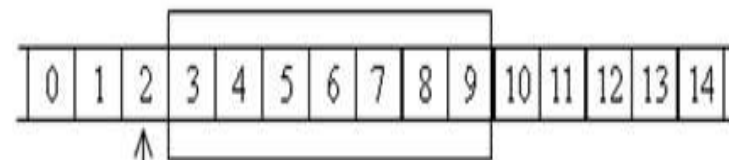
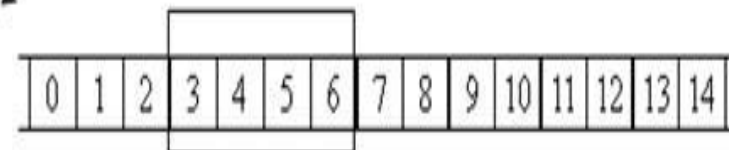


LAR

Receiver

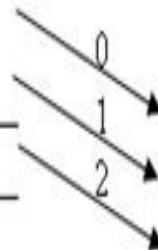


LAF



LFR

A2



Error Control

- Error detection—Receiver uses redundant information to detect whether the received data contains error or not. If an error is detected, the data is discarded and sender retransmits.
- Methods are Two dimensional parity, Internet checksum, CRC

Two-Dimensional Parity

- Data is divided into rows of 7 bits each.
- Even parity is computed for each row (Vertical Redundancy Check) and each column (Longitudinal Redundancy Check).
- Thus a parity byte for the entire frame, in addition to a parity bit for each byte is sent.
- Receiver computes row and column parities for data bits. If all parity bits and parity byte match, then the frame is accepted else discarded.
- Two-dimensional parity catches all 1, 2 and 3-bit errors, and most 4-bit errors.

Data : 1100111 1011101 0111001 0101001

1 1 0 0 1 1 1 1

1 0 1 1 1 0 1 1

0 1 1 1 0 0 1 0

0 1 0 1 0 0 1 1

0 1 0 1 0 0 1 1 Column Parities

Internet Checksum

Sender

- Given data is divided into 16-bit words.
- All words are added using one's complement arithmetic. Carries (if any) are wrapped and added to the sum. Complement of sum is known as *checksum* and is sent with data

Receiver

- All words including checksum are added using one's complement addition.
- The sum is complemented and becomes the *new checksum*. If the value of checksum is 0, the message is accepted, otherwise it is rejected

Cyclic Redundancy Check (CRC)

- CRC uses the concept of finite fields.
- A n bit message is represented as a polynomial of degree $n - 1$.
- Message $M(x)$ is represented as a polynomial by using the value of each bit as coefficient for each term. For example, 10011001 is represented as $x^7 + x^4 + x^3 + 1$
- For calculating a CRC, sender and receiver agree on a divisor polynomial, $C(x)$ of degree k such that $k \leq n - 1$

Sender

- Multiply $M(x)$ by x^k i.e., append k zeroes. Let the modified polynomial be $M'(x)$
- Divide $M'(x)$ by $C(x)$ using XOR operation. The remainder has k bits
- Subtract the remainder from $M'(x)$ using XOR, say $T(x)$ and transmit $T(x)$ with $n + k$ bits

Receiver

- Divide the received polynomial $T'(x)$ by $C(x)$ as done in sender
- If the remainder is non-zero then discard the frame
- If zero, then no errors and redundant bits are removed to obtain data

Media Access Control

- If access to medium is not regulated, then multiple stations may transmit simultaneously resulting in collisions and retransmission.
- Access to the medium could be either random or controlled or channelized.

<i>Random access</i>	<i>Controlled access</i>	<i>Channelization</i>
ALOHA	RESERVATION	FDMA
CSMA – CSMA/CA CSMA/CD	POLLING	TDMA
	TOKEN PASSING	CDMA

RANDOM ACCESS- ALOHA

- ALOHA is a random access protocol. Station don't sense the medium before transmitting
- Resulted in high amount of collision with 18.4% throughput.
- When stations transmitted only at beginning of time slots, throughput doubled to 36.8%

RANDOM ACCESS- CSMA

- In Carrier Sense Multiple Access (CSMA), each station first checks state of the link.
- Nodes using CSMA can distinguish between an idle and busy link.
- If link is idle, then data is sent, else attempted after waiting random amount of time.
- Two or more nodes may find the medium idle and transmit. Thus collision still exists.

CSMA- CD

- CSMA with collision detection (CSMA/CD) handles collisions over a wired medium.
- Station monitors the medium as it sends a frame
- If a collision is detected, it aborts transmission and broadcasts a jamming signal.
- Stations hearing jamming signal, refrain from transmitting frames.
- It waits for a random amount of time and attempts retransmission.

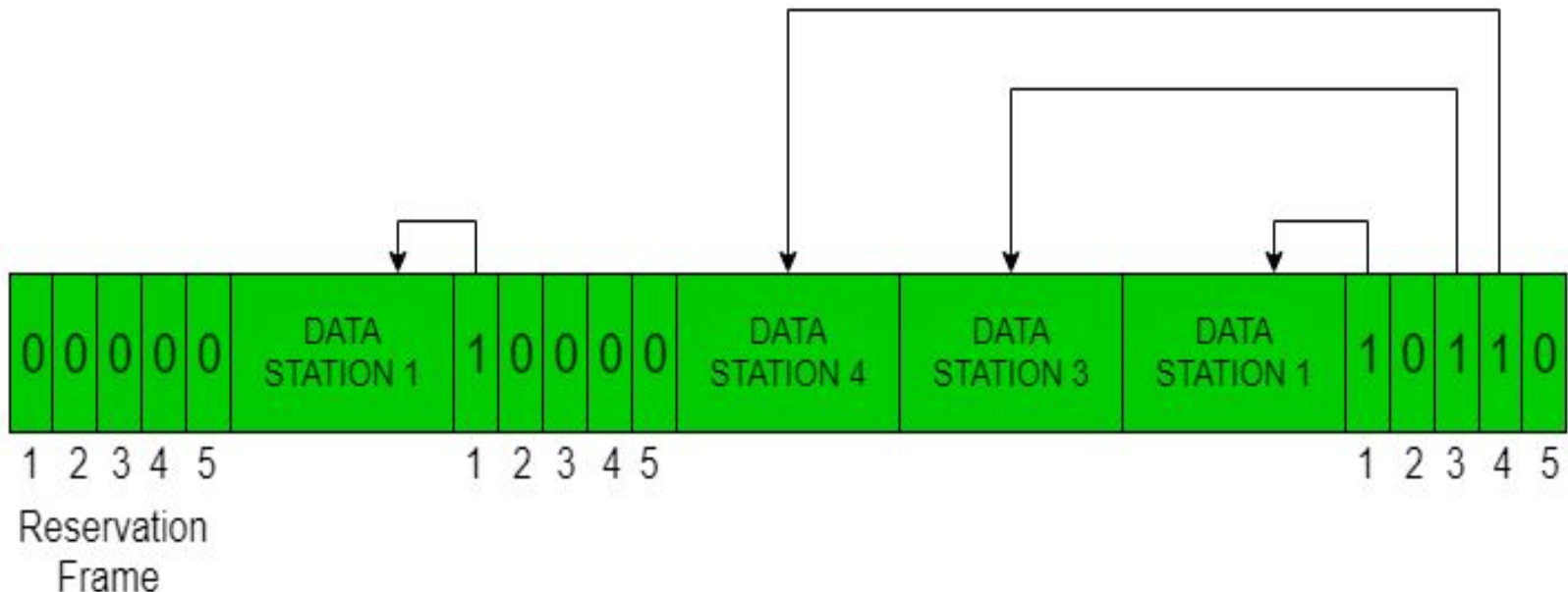
CONTROLLED ACCESS

- In controlled access, the stations seek data from one another to find which station has the right to send. It allows only one node to send at a time, to avoid the collision of messages on a shared medium. The three controlled-access methods are:
 - Reservation
 - Polling
 - Token Passing

RESERVATION

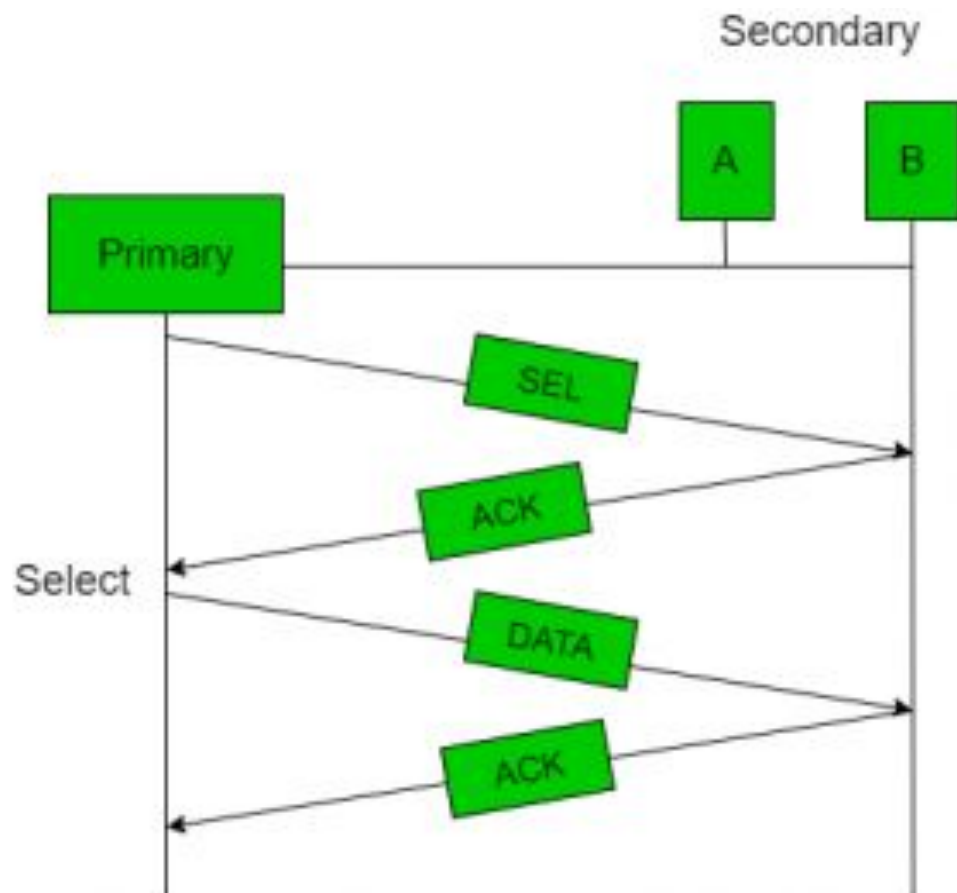
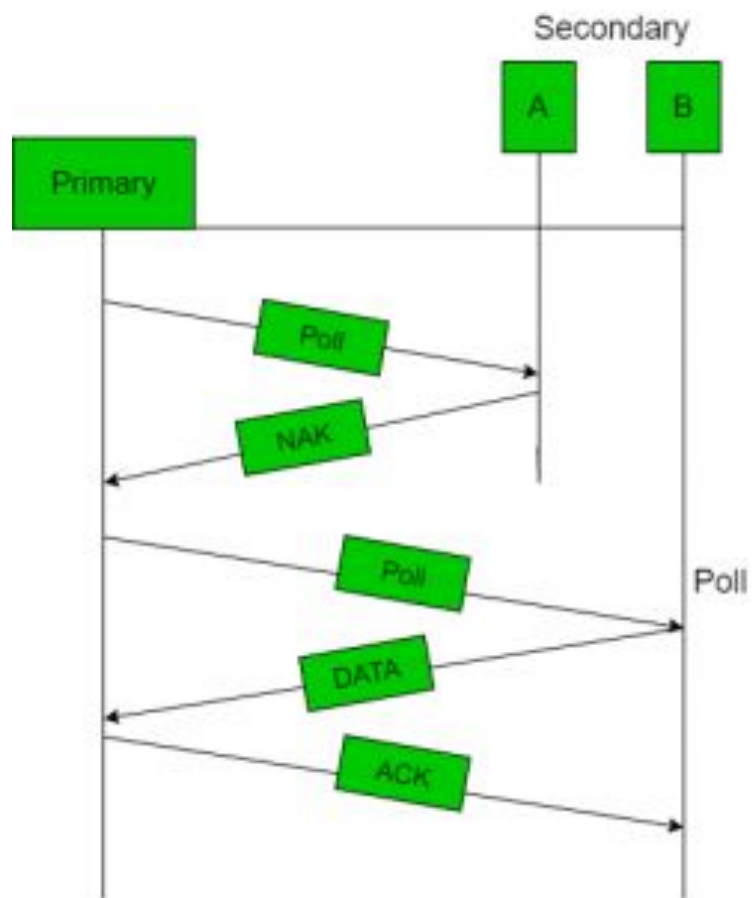
- In the reservation method, a station needs to make a reservation before sending data.
- The timeline has two kinds of periods:
 - Reservation interval of fixed time length
 - Data transmission period of variable frames.
- If there are M stations, the reservation interval is divided into M slots, and each station has one slot.
- Suppose if station 1 has a frame to send, it transmits 1 bit during the slot 1. No other station is allowed to transmit during this slot.
- In general, i^{th} station may announce that it has a frame to send by inserting a 1 bit into i^{th} slot. After all N slots have been checked, each station knows which stations wish to transmit.

- The stations which have reserved their slots transfer their frames in that order.
- After data transmission period, next reservation interval begins.
- Since everyone agrees on who goes next, there will never be any collisions.



POLLING

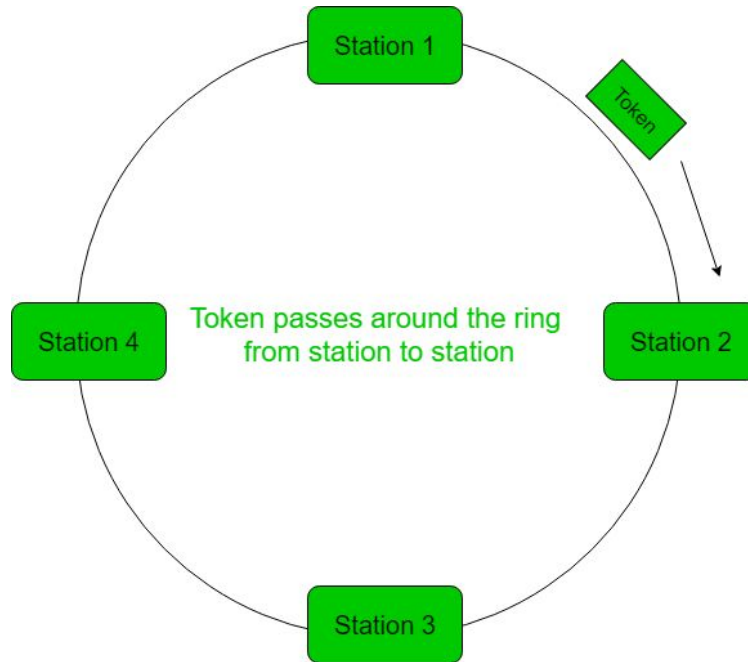
- Polling process is similar to the roll-call performed in class. Just like the teacher, a controller sends a message to each node in turn.
- In this, one acts as a primary station(controller) and the others are secondary stations. All data exchanges must be made through the controller.
- The message sent by the controller contains the address of the node being selected for granting access.
- Although all nodes receive the message the addressed one responds to it and sends data if any. If there is no data, usually a “poll reject”(NAK) message is sent back.
- Problems include high overhead of the polling messages and high dependence on the reliability of the controller.



TOKEN PASSING

- In token passing scheme, the stations are connected logically to each other in form of ring and access to stations is governed by tokens.
- A token is a special bit pattern or a small message, which circulate from one station to the next in some predefined order.
- In Token ring, token is passed from one station to another adjacent station in the ring whereas incase of Token bus, each station uses the bus to send the token to the next station in some predefined order.
- In both cases, token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station. If it has no queued frame, it passes the token simply.

- After sending a frame, each station must wait for all N stations (including itself) to send the token to their neighbours and the other $N - 1$ stations to send a frame, if they have one.
- There exists problems like duplication of token or token is lost or insertion of new station, removal of a station, which need be tackled for correct and reliable operation of this scheme.



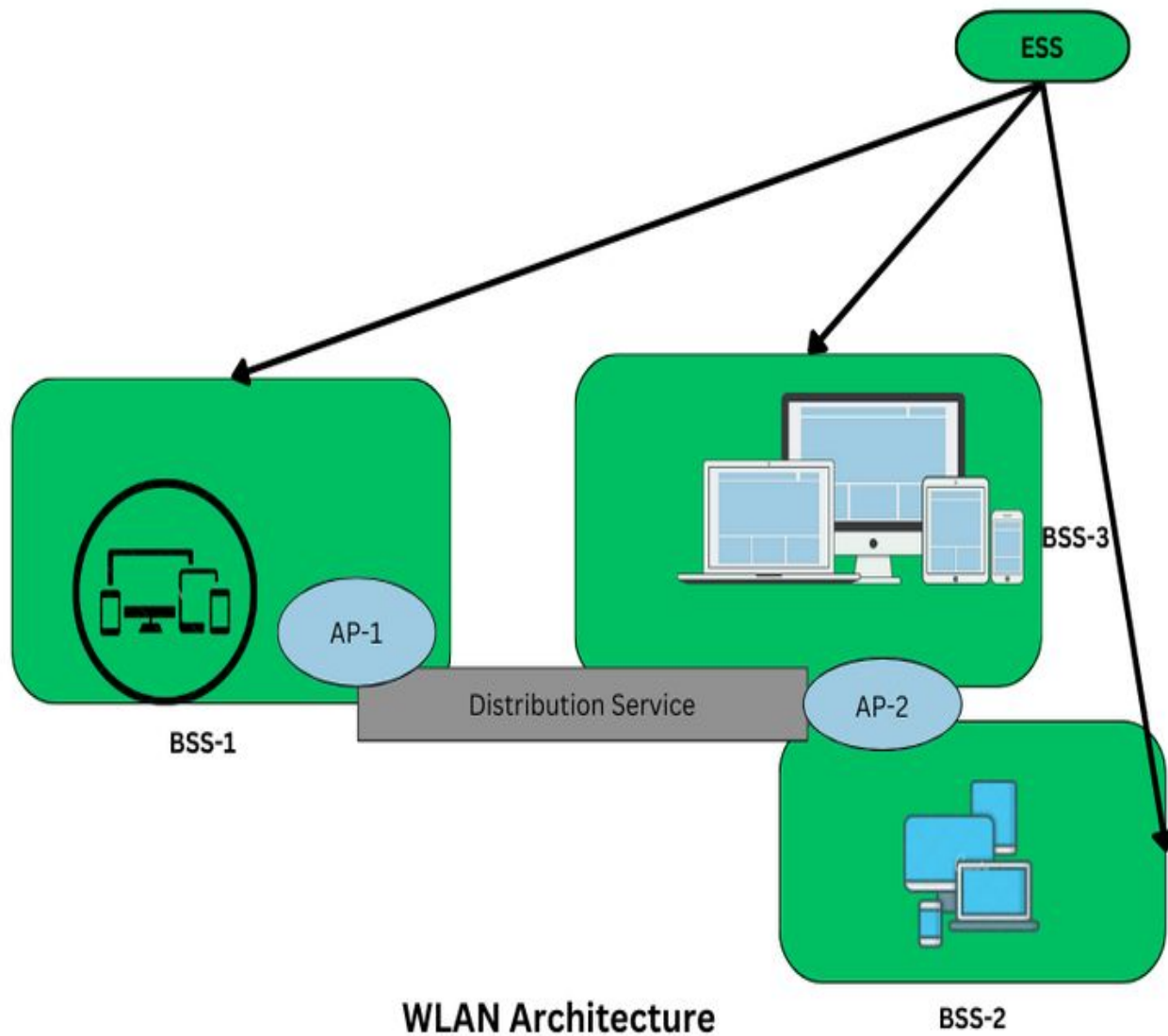
Wireless LAN –IEEE 802.11

A WLAN basically, extends a wired local area network. WLAN's are built by attaching a device called the access point(AP) to the edge of the wired network.

Architecture:

Components in Wireless LAN architecture as per IEEE standards are as follows:

- Stations:** Stations consist of all the equipment that is used to connect all wireless LANs. Each station has a wireless network controller.
- Base Service Set(BSS):** It is a group of stations communicating at the physical layer.
- Extended Service Set(ESS):** It is a group of connected Base Service Set(BSS).
- Distribution Service (DS):** It connects all Extended Service Set(ESS).

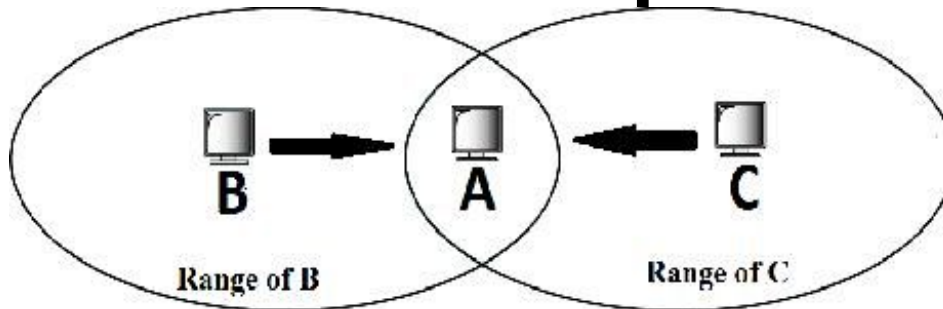


Types of WLANs

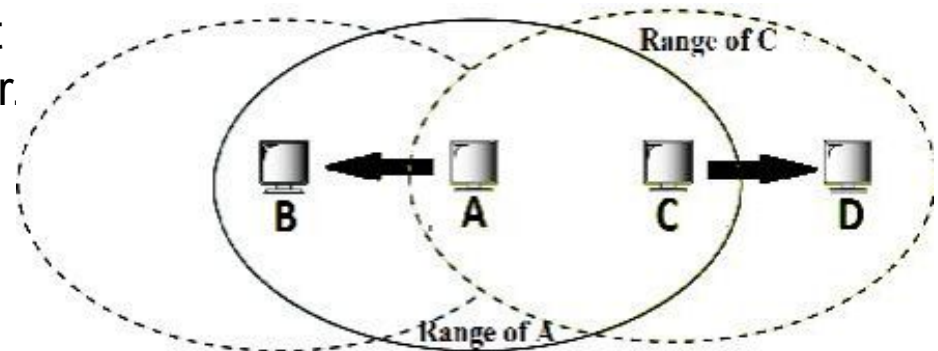
As per IEEE standard WLAN is categorized into two basic modes, which are as follows:

- **Infrastructure:** In Infrastructure mode, all the endpoints are connected to a base station and communicate through that; and this can also enable internet access.
- A WLAN infrastructure can be set up with: a wireless router (base station) and an endpoint (computer, mobile phone, etc). An office or home WiFi connection is an example of Infrastructure mode.
- **Ad Hoc:** In Ad Hoc mode WLAN connects devices without a base station, like a computer workstation. An Ad Hoc WLAN is easy to set up it provides peer-to-peer communication. It requires two or more endpoints with built-in radio transmission.

Hidden / Exposed Node Problem



- *Hidden Node* : Suppose node *B* is sending data to *A*. At the same time, node *C* also wishes to send to *A*. Since node *B* is not within the range of *C*, *C* finds the medium free and transmits to *A*.
- Frames from nodes *B* and *C* sent and *C* are *hidden* from each other.



- Suppose node *A* is transmitting to node *B* and node *C* has some data to be sent to node *D*. Node *C* finds the medium busy, since it hears the transmission from node *A* and refrains from sending to node *D*, even though its transmission to *D* would not interfere.
- Thus node *C* is *exposed* to transmission from node *A* to *B*

Access Control: CSMA/CA: **Multiple Access with Collision Avoidance (MACA)**

- Nodes exchange *control frames* to reserve access, so that nearby nodes avoid transmission during duration of a data frame.
- Control frames used to avoid collision are *Request to Send* (RTS) and *Clear to Send* (CTS).
- Both RTS and CTS contain source & destination address and transmission duration.
- Source sends RTS frame to destination
- Nodes that receive RTS frame are close to source and wait for CTS to be transmitted back.
- Destination node sends back a CTS frame as acknowledgement.
- Nodes that receive CTS remain silent for the upcoming data transmission.
- Nodes that receive RTS but not CTS, is away from the destination and is free to transmit.
- Destination sends an ACK frame finally after successfully receiving data frames.

Frame Format



Control—indicates frame type (RTS, CTS, ACK or data) and 1-bit ToDS / FromDS

□ *Duration*—specifies duration of frame transmission.

□ *Addresses*—The *four* address fields depend on value of ToDS and FromDS subfields.

□ *Sequence Control*—defines sequence number of the frame.

□ *Body*—contains a maximum of 0–2312 bytes.

□ *CRC*—contains CRC-32 error detection sequence.

Addressing

To DS	From DS	Addr 1	Addr 2	Addr 3	Addr 4	Description
0	0	Destination	Source			Send Directly
0	1	Destination	Sending AP	Source		Frame come from DS
1	0	Receiving AP	Source	Destination		Frame goes to DS
1	1	Receiving AP	Sending AP	Destination	Source	Frame goes from one AP to another.

Bluetooth

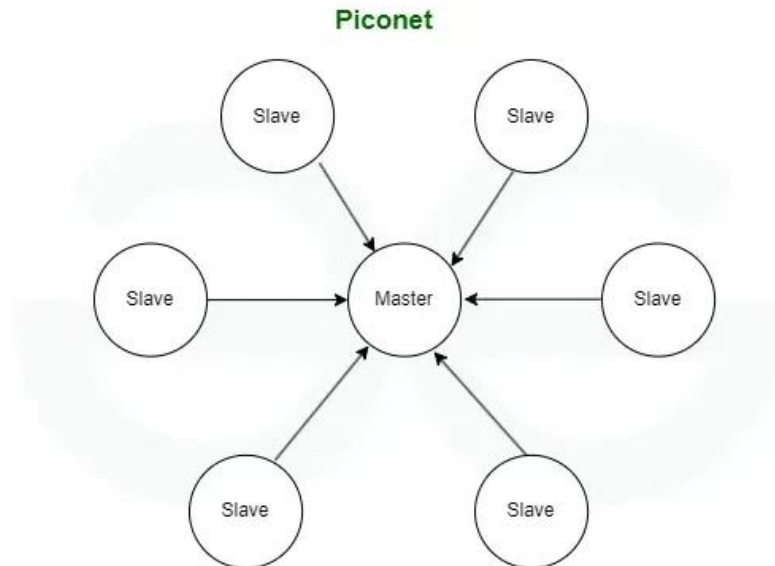
- WPAN- used for data communication over smaller distances.
- a wireless technology that lets devices like phones, tablets, and headphones connect to each other and share information without needing cables. Bluetooth simply follows the principle of transmitting and receiving data using radio waves.
- FHSS, 10 meters , 1 to 3 Mbps.
- When two devices start to share data, they form a network called piconet which can further accommodate more than five devices.

Architecture of Bluetooth

Piconet

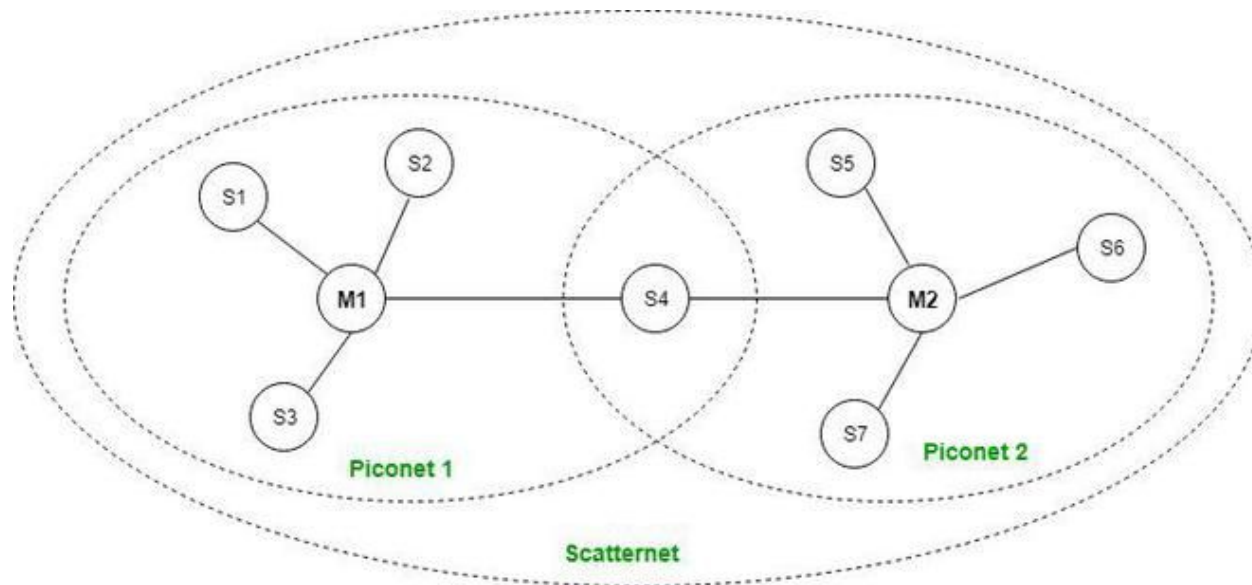
Piconet is a type of Bluetooth network that contains one primary node called the master node and seven active secondary nodes called slave nodes, that there is a total of 8 active nodes which are present at a distance of 10 meters.

The communication between the primary and secondary nodes can be one-to-one or one-to-many.

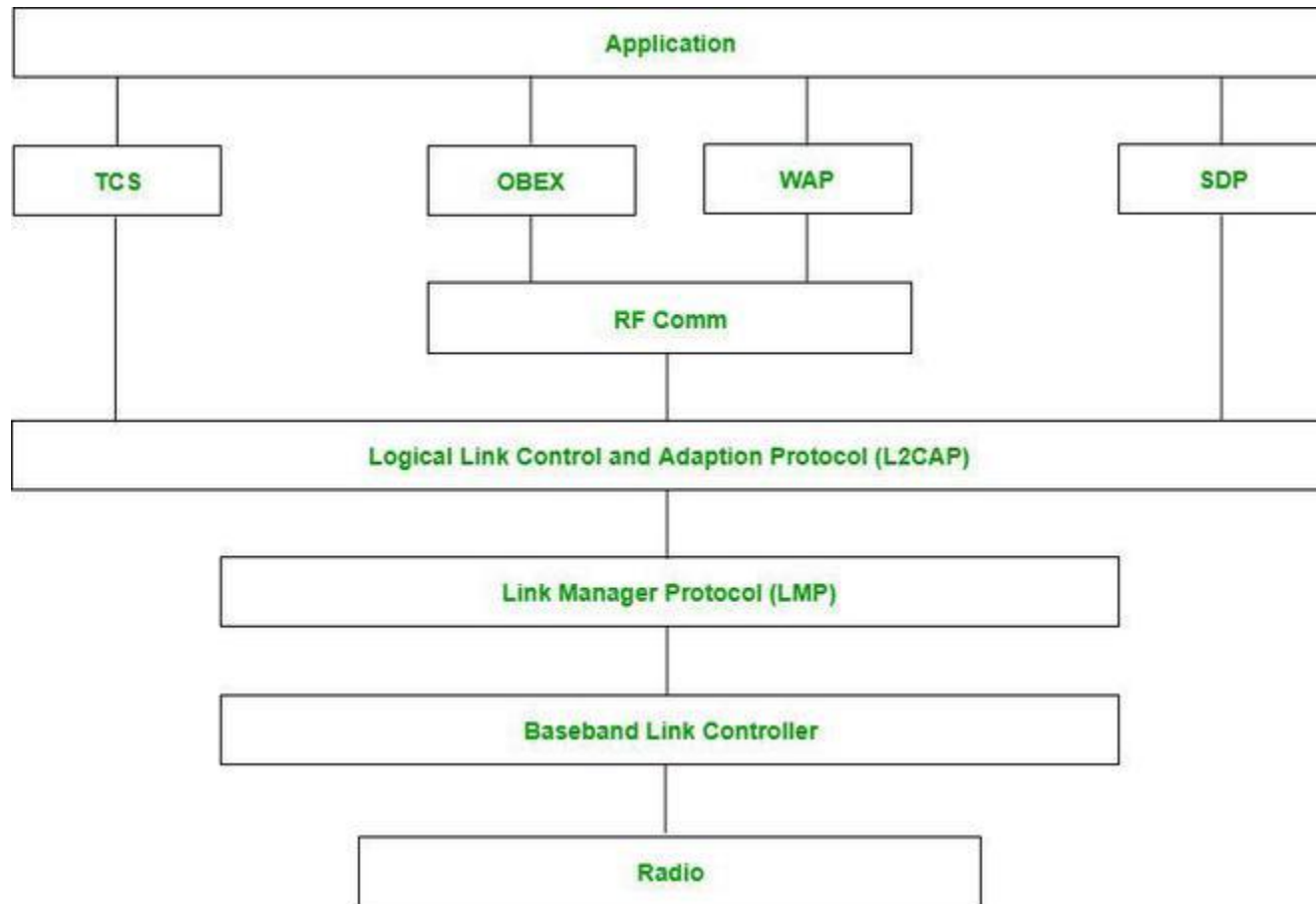


Scatternet

- It is formed by using various piconets. A slave that is present in one piconet can act as master or we can say primary in another piconet.
- This kind of node can receive a message from a master in one piconet and deliver the message to its slave in the other piconet where it is acting as a master.
- This type of node is referred to as a bridge node. A station cannot be mastered in two piconets.



Protocol Stack



- **Radio (RF) Layer:** It specifies the details of the air interface, including frequency, the use of frequency hopping and transmit power. It performs modulation/demodulation of the data into RF signals. It defines the physical characteristics of Bluetooth transceivers. It defines two types of physical links: connection-less and connection-oriented.
- **Baseband Link Layer:** The baseband is the digital engine of a Bluetooth system and is equivalent to the MAC sublayer in LANs. It performs the connection establishment within a piconet, addressing, packet format, timing and power control.
- **Link Manager Protocol Layer:** It performs the management of the already established links which includes authentication and encryption processes. It is responsible for creating the links, monitoring their health, and terminating them gracefully upon command or failure.
- **Logical Link Control and Adaption (L2CAP) Protocol Layer:** It is also known as the heart of the Bluetooth protocol stack. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs segmentation and multiplexing.

- **Service Discovery Protocol (SDP) Layer:** It is short for Service Discovery Protocol. It allows discovering the services available on another Bluetooth-enabled device.
- **RF Comm Layer:** It is a cabal replacement protocol. It is short for Radio Frontend Component. It provides a serial interface with WAP and OBEX. It also provides emulation of serial ports over the logical link control and adaption protocol(L2CAP). The protocol is based on the ETSI standard TS 07.10.
- **OBEX:** It is short for Object Exchange. It is a communication protocol to exchange objects between 2 devices.
- **WAP:** It is short for Wireless Access Protocol. It is used for internet access.
- **TCS:** It is short for Telephony Control Protocol. It provides telephony service. The basic function of this layer is call control (setup & release) and group management for the gateway serving multiple devices.
- **Application Layer:** It enables the user to interact with the application.

Types and Applications

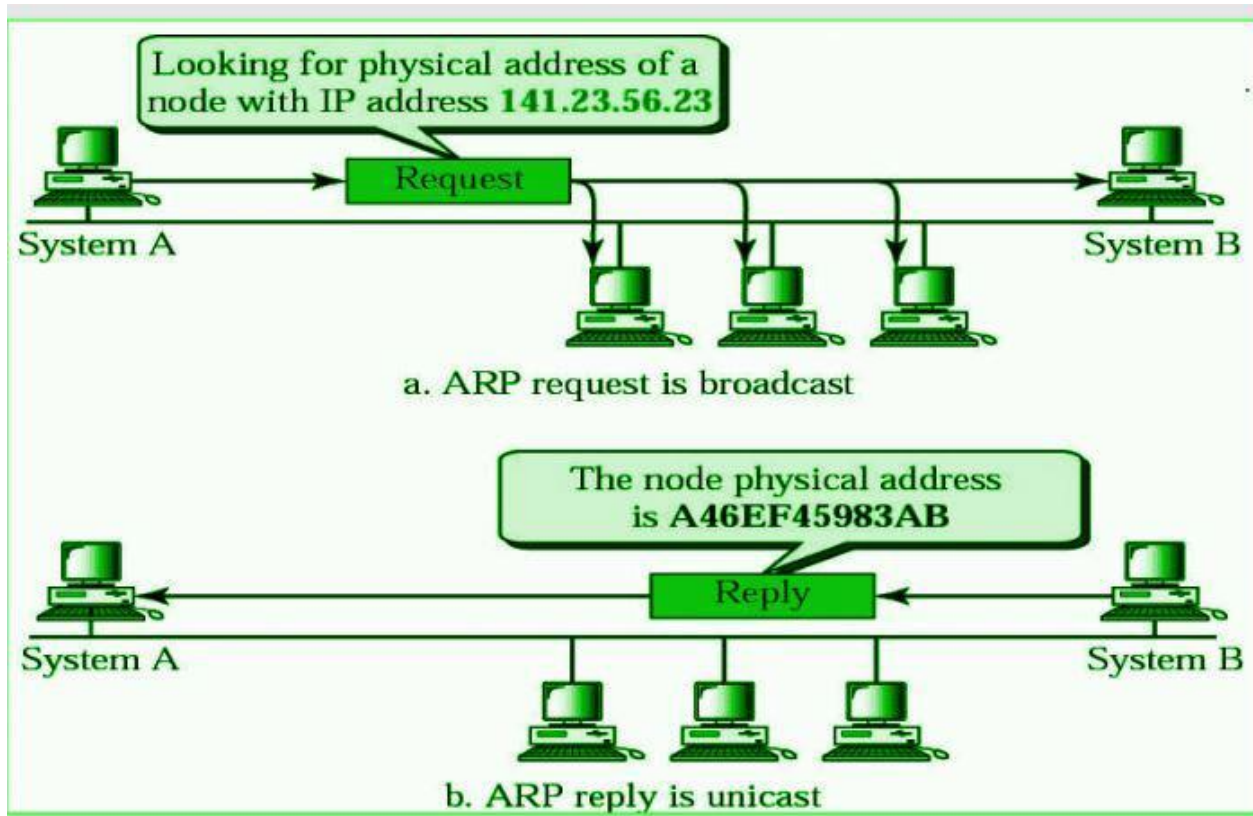
- It is a low-cost and easy-to-use device.
- It can also penetrate through walls.
- It creates an Ad-hoc connection immediately without any wires.
- It is used for voice and data transfer.

Types:

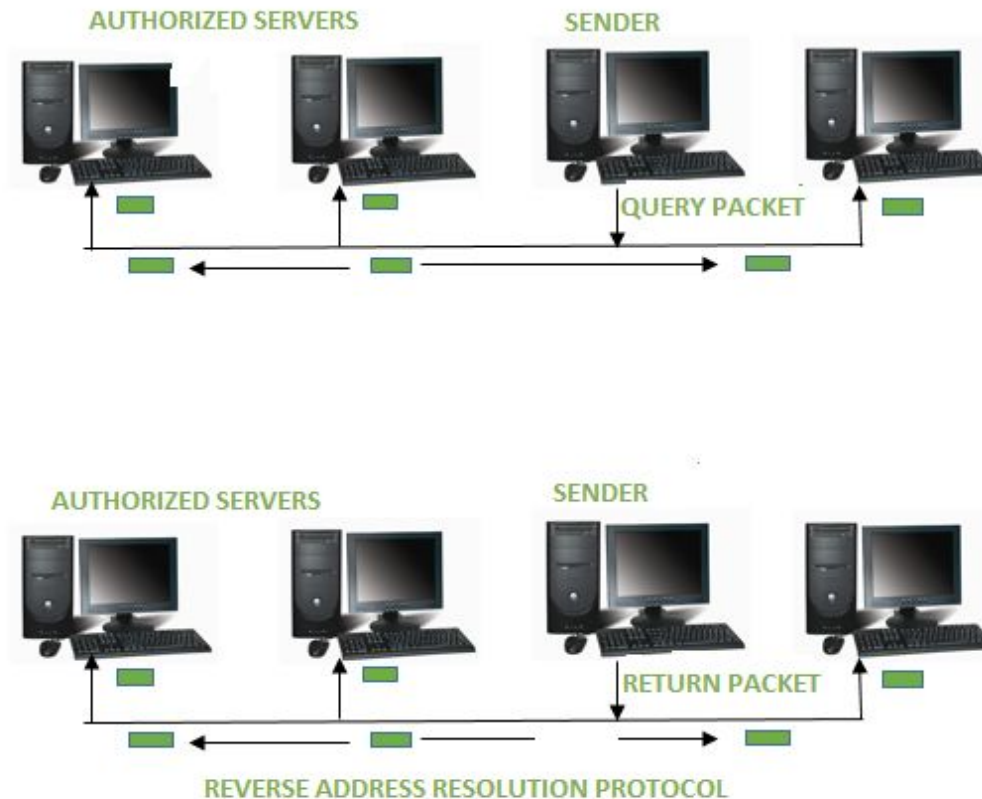
- **In-Car Headset and Stereo Headset:** One can make calls from the car speaker system without the use of mobile phones and To listen to music in car or in music players at home.
- **Webcam:** One can link the camera with the help of Bluetooth with their laptop or phone.
- **Bluetooth-Equipped Printer:** The printer can be used when connected via Bluetooth with mobile phone or laptop.
- **Bluetooth Global Positioning System (GPS):** To use Global Positioning System (GPS) in cars, one can connect their phone with car system via Bluetooth to fetch the directions of the address.

Related Protocols

(1) **ARP**: is a protocol used to map an IP address 32-bit to a physical **MAC address 48-bit**. The MAC address is known as the hardware id number.



(ii) RARP: is used to map a MAC address 48-bit to an IP address 32-bit. This protocol is typically used by devices that know their **Media Access Control address** but need to find their IP address.



Difference between ARP and RARP

ARP	RARP
A protocol used to map an IP address to a physical address	A protocol used to map a physical address to an IP address
To obtain the MAC address of a network device when only its IP address is known	To obtain the IP address of a network device when only its MAC address is known
IP addresses	MAC addresses
ARP stands for Address Resolution Protocol.	Whereas RARP stands for Reverse Address Resolution Protocol.
In ARP, broadcast MAC address is used.	While in RARP, broadcast IP address is used.
In ARP, ARP table is managed or maintained by local host .	While in RARP, RARP table is managed or maintained by RARP server.
In Address Resolution Protocol, Receiver's MAC address is fetched.	While in RARP, IP address is fetched.
ARP is used in sender's side to map the receiver's MAC address.	RARP is used in receiver's side to map the sender's IP.

(iii) NAT (Network Address Translation)

- **Network Address Translation (NAT)** is a process in which one or more local IP addresses are translated into one or more Global IP addresses and vice versa to provide Internet access to the local hosts.
- It also does the translation of port numbers, i.e., masks the port number of the host with another port number in the packet that will be routed to the destination.
- It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on a router or firewall.



ICMP

- ICMP is used for reporting errors and management queries. It is a supporting protocol and is used by network devices like routers for sending error messages and operations information. For example, the requested service is not available or a host or router could not be reached.
- Another important use of ICMP protocol is used to perform network diagnosis by making use of traceroute and ping utility.

- ☐ Definition
- ☐ usage
- ☐ frame format
- ☐ DDOS attacks
- ☐ types

ICMP Frame format

Type(8 bit)	Code(8 bit)	Checksum(16 bit)
Extended Header(32 bit)		
Data/Payload(Variable Length)		

ICMP in DDOS Attacks

- **Ping of Death attack:**

Whenever an attacker sends a ping, whose size is greater than the maximum allowable size, oversized packets are broken into smaller parts. When the sender re-assembles it, the size exceeds the limit which causes a buffer overflow and makes the machine freeze. This is simply called a Ping of Death Attack.

- **ICMP Flood Attack:**

Whenever the sender sends so many pings that the device on whom the target is done is unable to handle the echo request. This type of attack is called an ICMP Flood Attack. This attack is also called a ping flood attack. It stops the target computer's resources and causes a denial of service for the target computer.

- **Smurf attack:**

Smurf Attack is a type of attack in which the attacker sends an ICMP packet with a spoofed source IP address. These type of attacks generally works on older devices like the ping of death attack.

Type	Code	Description
0 - Echo Reply	0	Echo reply
3 - Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation is needed and the DF flag set
	5	Source route failed
5 - Redirect Message	0	Redirect the datagram for the network
	1	Redirect datagram for the host
	2	Redirect the datagram for the Type of Service and Network
	3	Redirect datagram for the Service and Host
8 - Echo Request	0	Echo request

9 - Router Advertisement	0	Use to discover the addresses of operational routers
10 - Router Solicitation	0	
11 - Time Exceeded	0	Time to live exceeded in transit
	1	Fragment reassembly time exceeded.
12 - Parameter Problem	0	The pointer indicates an error.
	1	Missing required option
	2	Bad length
13 - Timestamp	0	Used for time synchronization
14 - Timestamp Reply	0	Reply to Timestamp message